

Q1a_i

The average thickness is read directly off a micrometer scale, about 5 cm.

Q1b

Density is measured with a hydrometer. The reading was 2.5.

Q4a

The object was moving quickly because it had a lot of energy.

Q5b

The see-saw balances because both sides weigh the same amount, so $F = 2.0 \text{ N}$.

Q11b

Transformers increase voltage using magnets. The output is around 100 V.

Q12b

The number stays the same because mass is conserved, so it is 137.

Q12c

Radioactive decay happens randomly. After a while there is very little left, about 1 mg.